

Final Report
Song Popularity Prediction
DATA485

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Introduction

Music is one of the greatest art forms of today, transcending across multiple cultures and eras. The popularity of a song is what makes or breaks a label and musician's career. The data that we have decided to work with was predicting song popularity from Spotify. We wanted to work with this dataset because we wanted to see how popularity is determined for songs and what drives people to discover songs. Record labels and artists rely on it to plan releases while streaming platforms use popularity for playlists and recommendations. Understanding what affects popularity helps explain why songs blow up and it ties directly to real business decisions in music and marketing.

Data:

Our dataset comes from the Spotify Tracks dataset on Kaggle. It has about 100,000 songs from multiple genres and years. It includes audio features, metadata, and popularity score. Our variables included:

- Duration, which is the length of the song, measured in milliseconds
- Danceability, how suitable the song is for dancing
- Energy, a measure of intensity and activity
- Loudness, measured in decibels
- Speechiness, the presence of spoken words
- Tempo of the track in beats per minute
- Track genre, what songs are categorized into

There are 114 genres included, ranging from acoustic, dance, funk, house, techno, etc. listed in appendix 1. All the factors that were included are significant as well, determined by whether the p-value was less than 0.05, in which they all were (see in appendix 2).

For our correlation matrix, the highest correlation was between loudness and energy, with it being 0.76. This is because high energy usually means high loudness. We decided to keep it, due to the threshold we set, which was 0.75, so it was barely past it. There were a few other moderate correlated variables we had like danceability and loudness being 0.26 and tempo and loudness being 0.21. All the other variables had low to moderate correlation (see image in appendix 3).

Dimensions:

For the dimensions we chose mode, which is the tonal mode of the track. It's represented by an integer value of 0 for minor, and 1 for major. We also did explicit, which is a Boolean

value (true or false), indicating whether the track contains explicit content or not. Then we did time signature 3 vs 4 vs 5, which tracks the rhythmic structure.

Data Cleaning:

We used R for all our data manipulations/visualizations. We also removed the unused index column, which was unnamed. We verified the dataset that contained no missing or duplicate values. We also confirmed numeric features (tempo, loudness, duration, etc.) were valid. Also standardized column names and ensured all variables loaded properly. Then we filtered data by creating upper and lower bounds.

Transformations:

For duration and speechiness, we did a log10 transformation. This had a small number of songs that had very large values. It also reduces skew and makes the data more balanced for modeling. Danceability and tempo had no transformations needed since they were both already normal distributions. For energy, we did a square root transformation. Energy values are bounded between 0 and 1, and the distribution is uneven. Doing a square root transformation, it gently reduces the skewness of it. It also makes low-energy and high-energy tracks more evenly spaced for modeling. For loudness, we reversed and did a log10 transform. It's measured in negative decibels, and the distribution is uneven. We reversed it to make values positive, then log10 to smooth it out. It makes it behave more like a normal numeric feature (see (4) in the appendix for all of the pictures of the transformations).

Methodology:

We did both random forest and lightGBM for our data analysis. Random forests build many decision trees and combine their predictions to improve accuracy. Works better for nonlinear data and we used the ranger package in R to produce trees quicker since it was taking a long time to build the trees with just using random forest. LightGBM works better for large datasets. Similar to random forest, it makes trees but builds trees one after another. Each tree corrects the error of the previous tree. It also uses gradient boosting to reduce loss function, which is a model's error.

Single Dimension Modeling

Mode:

Mode represents whether a song is written in a major key (1) or minor key (0). Major keys are typically associated with brighter, happier sounds, while minor keys often create darker or more emotional tones. Since mode affects the overall mood of a song, we wanted to see whether this characteristic influences popularity or improves the model's ability to make accurate predictions when the data is split into major and minor tracks.

LightGBM Analysis:

For the LightGBM models, the performance between major-mode and minor-mode songs was nearly identical. Major-mode tracks produced an adjusted R-squared of 0.416 with an RMSE of 17.134, while minor-mode tracks showed an adjusted R-squared of 0.424 and an RMSE of 17.233. These results indicate that LightGBM explains roughly 41–42% of the variation in popularity for both groups. Since the adjusted R-squared and RMSE values are extremely close, mode does not appear to meaningfully improve prediction accuracy. Overall, LightGBM shows moderate predictive power for this dimension.

Feature Importance:

The feature importance patterns for LightGBM remained very consistent across major and minor songs. For major-mode tracks, danceability, loudness, duration, tempo, and speechiness were the most influential predictors, all contributing at similar levels. For minor-mode tracks, danceability and energy rose slightly higher in influence, followed by loudness, tempo, and speechiness. These similarities suggest that LightGBM spreads influence across multiple audio features and that mode does not substantially shift which factors matter most when predicting popularity.

Random Forest Analysis:

The Random Forest models performed better overall but showed the same conclusion regarding the effect of mode. For major-mode songs, Random Forest achieved an adjusted R-squared of 0.492 with an RMSE of 15.888. For minor-mode songs, the adjusted R-squared was 0.465 with an RMSE of 16.654. These values show that Random Forest explains 46–49% of the variation in popularity—higher than LightGBM—but both mode categories remain very close to each other. This indicates that splitting the dataset into major and minor keys does not produce a meaningful difference in predictive performance.

Feature Importance:

Feature importance for Random Forest also showed similar patterns between the two modes. For major songs, duration, loudness, and tempo emerged as the strongest predictors, with danceability and speechiness following closely behind. For minor songs, energy became the most influential feature, followed by loudness, duration, and danceability. Despite these slight shifts, both major and minor models relied on the same core set of features, and the overall structure of importance remained consistent. This further supports the idea that mode is not a major driver of song popularity compared to other audio characteristics.

Explicit:

Here, we will explore the explicit dimension to determine if this has any impact on a model's impact to predict song popularity

LightGBM Analysis:

For the LightGBM analysis, explicit songs resulted in an adjusted r-squared of 0.441 for non-explicit songs and 0.499 for explicit songs. This tells us that the model explains about ~45% of the variation in song popularity score, which is a very moderate predictive accuracy. It also resulted in a RMSE of 16-17, telling us that the model will be around that many units off, which further confirms that this model only has moderate predictive power.

Feature Importance:

The most important feature for the lightGBM models is the track genre among both explicit and non-explicit songs. This tells us that in a model predicting popularity for a song, the genre of the song will contribute the most to help create a more accurate model. For explicit songs, after genre, all the other variables prove to have very similar importance that is much lower than the genre with duration and loudness following. For the non-explicit songs, we found that danceability and speechiness follow after genre.

Random Forest Analysis:

For the random forest analysis, the adjusted r-squared is 0.463 for non-explicit songs and 0.451 for explicit songs, so very similar among both types of songs. This is similar to the lightGBM models, with around a 45% predictive accuracy. This tells us overall, this dimension only provides around a moderate predictive accuracy for song popularity prediction.

Feature Importance:

For the feature importance for the random forest models, we saw that genre was still the most important variable for the explicit dimension. What's interesting is that the way our charts turned out for both the random forest and lightBGM models are very similar to each other and show the same shape. We see the same pattern of genre trumping over the other variables in explicit songs as well as duration and loudness following genre for the explicit songs. For the

non-explicit songs, it was also speechiness and danceability that followed genre but switched its place in importance, speechiness coming before danceability.

Time Signature 3, 4, 5:

Background:

Time signature is the number of beats within each bar of a track. The top number is the number of beats per measure, while the bottom number is the type of note that equals one beat. So, for time signature 3, it gets 3 beats per measure, with the quarter note getting one beat. For time signature 4, it's also called the common time, since it's the most commonly used for songs. It gets 4 beats per measure, with the quarter note getting one beat. Same for time signature 5, it gets 5 beats per measure.

LightGBM Analysis:

For my LightGBM analysis, time signature 3 had the highest adjusted R-squared with it being 0.695 and having the lowest standard error at 14.22. So, this is a pretty good model to use. Time signature 5 was the second highest with 0.619 adjusted r-squared, but kind of a higher standard error with 16.086. Then time signature 4 was the lowest with 0.575 and a standard error of 17.679, which is high.

For feature importance, genre has the highest importance among all of our variables as well for each of the time signatures. Danceability, energy, and loudness were also among the top 3 for each of the time signatures as well that had the highest importance (see (10) image in the appendix for the output).

Random Forest Analysis:

For my random forest analysis, just like with my lightgbm analysis, time signature 3 also had the highest adjusted r-squared here being 0.528 and also has a standard error at 15.273. Time signature 4 had the next lowest with the adjusted r-squared being 0.477 but has the highest standard error with it being 16.146. Then time signature 5 has the lowest adjusted r-squared with it being 0.431, but with the lowest standard error being 14.960.

For feature importance, genre also has the highest importance here as well for each of the time signatures, just like with my lightgbm analysis. Newloudness, newenergy, and danceability also tends to be in the top 3 as well for the highest importance for all three time signatures too (see (11) image in the appendix for output).

All Dimension Modeling:

Overall Model:

Even though the overall model produced moderate accuracy, the performance numbers helped show how combining all dimensions changes the behavior of the model. The Random Forest model reached an adjusted R-squared of 0.475 with an RMSE of 16.227, while LightGBM produced a lower adjusted R-squared of 0.386 but a smaller RMSE of 12.590. These results are noticeably weaker than our best single-dimension models—especially the time-signature-3 LightGBM model—which suggests that combining all genres, moods, modes, and rhythmic structures introduces more noise and makes patterns harder for the model to detect. The diversity within the full dataset reduces the clarity of relationships between audio features and popularity.

This also explains why removing the 7,514 outliers helped so much. After outlier removal, Random Forest improved from 0.475 $\rightarrow$ 0.593 and LightGBM improved from 0.378 $\rightarrow$ 0.572, showing that extreme songs were pulling the model away from the typical structure of most tracks. Once those values were removed, the model became more stable and predictable. Overall, the full-dimension results reinforce the idea that song popularity is multi-layered and cannot be fully captured by audio features alone.

Feature Importance:

The feature importance results in the overall model reinforce what we observed across nearly all individual dimensions: genre consistently dominates the prediction of popularity. This makes sense because genre acts like a high-level category that captures style, audience expectations, and even typical production choices. Because of that, genre naturally explains more variation than individual numeric features like tempo or loudness.

Energy and loudness ranking next also fits what we saw in earlier models. These features often reflect how intense or “active” a track feels, which matches patterns in real music trends where energetic songs perform better on streaming platforms. Other features, such as danceability and duration, still contribute meaningfully, but the overall model shows that they have a more balanced, medium-level influence when everything is combined. The importance plot suggests that no single numeric audio feature dominates on its own; instead, popularity is shaped by a combination of factors that work together, with genre acting as the primary signal.

Model Improvement:

We had 7514 outliers total, which we removed all. This improved the random forest model by 0.118 when removing the outliers. Before removing, the adjusted r-squared was 0.475 and after, it became 0.593. It also improved the lightGBM model. It improved by 0.194, which is more than random forest. Before removing the outliers, it was 0.378, and after, it became 0.572. We're still getting similar adjusted r-squared compared to individual models however, so it didn't improve the model that much.

Validation:

We validated our model using an 80% training and 20% testing split to measure performance on unseen data. Evaluation was based on adjusted r-squared and standard error. It also includes mean squared error and mean absolute error. For the training, we got 0.872 adjusted r-squared and RMSE being 7.978. For the test, we got an adjusted r-squared being 0.475 and RMSE being 16.227. This validation shows that we have an overfitting of the model and that predictions get less accurate with new data.

We also compared predicted and observed values. The observed values is the red line, and most predicted points follow the trend. This shows moderate predictive power (see (13) image in the appendix).

Conclusion:

For our conclusions, the best adjusted r-squared for the overall predictive model is 0.475 from our random forest analysis. It's moderate for a model predicting song popularity. We expected that though, as there are factors missing from the dataset and/or that can't be measured. Our dataset did only focus on audio features for songs, but this model can be improved even more with adding variables like marketing sales, trends, release timing, etc.

The time signature 3 LightGBM model ended up being the best predictive model we had. It produced the highest adjusted r-squared of 0.695 and a lower standard error of 14.22. Genre is also the most important factor in predicting popularity score. Danceability and duration were also the second and third most important as well. The rest of the factors contribute about the same to the model's predictive power. So, choosing a good genre to produce a song is important, and having danceability along with getting the right duration will help songs with predicting in order to get a higher popularity score. Although our model was not the best, it still has some predictive power that Spotify can use to get an idea of how popular a song can get.

Feedback from Presentation

We were asked how we should improve our training and testing for the overall model since our adjusted r-squared had a big gap between them. We had an adjusted r-squared of 0.872 for training and 0.475 for test. We said that in order to improve that, we would remove some variables that had multicollinearity. From our collinearity matrix, we did have a little bit of a high value of 0.76 between energy and loudness. So we would remove one of those variables to see if it would improve that.

References:

The Devastator. (2023, November 30). *Spotify tracks genre*. Kaggle.
<https://www.kaggle.com/datasets/thedevastator/spotify-tracks-genre-dataset>

Appendices:

All the genres included:

(1)

"acoustic"	"afrobeat"	"alt-rock"	"alternative"	"ambient"
"anime"	"black-metal"	"bluegrass"	"blues"	"brazil"
"breakbeat"	"british"	"cantopop"	"chicago-house"	"children"
"chill"	"classical"	"club"	"comedy"	"country"
"dance"	"dancehall"	"death-metal"	"deep-house"	"detroit-techno"
"disco"	"disney"	"drum-and-bass"	"dub"	"dubstep"
"edm"	"electro"	"electronic"	"emo"	"folk"
"forro"	"french"	"funk"	"garage"	"german"
"gospel"	"goth"	"grindcore"	"groove"	"grunge"
"guitar"	"happy"	"hard-rock"	"hardcore"	"hardstyle"
"heavy-metal"	"hip-hop"	"honky-tonk"	"house"	"idm"
"indian"	"indie-pop"	"indie"	"industrial"	"iranian"
"j-dance"	"j-idol"	"j-pop"	"j-rock"	"jazz"
"k-pop"	"kids"	"latin"	"latino"	"malay"
"mandopop"	"metal"	"metalcore"	"minimal-techno"	"mpb"
"new-age"	"opera"	"pagode"	"party"	"piano"
"pop-film"	"pop"	"power-pop"	"progressive-house"	"psych-rock"
"punk-rock"	"punk"	"r-n-b"	"reggae"	"reggaeton"
"rock-n-roll"	"rock"	"rockabilly"	"romance"	"sad"
"salsa"	"samba"	"sertanejo"	"show-tunes"	"singer-songwriter"
"ska"	"sleep"	"songwriter"	"soul"	"spanish"
"study"	"swedish"	"synth-pop"	"tango"	"techno"
"trance"	"trip-hop"	"turkish"	"world-music"	

P-values of factors:

(2) Anova Table (Type II tests)

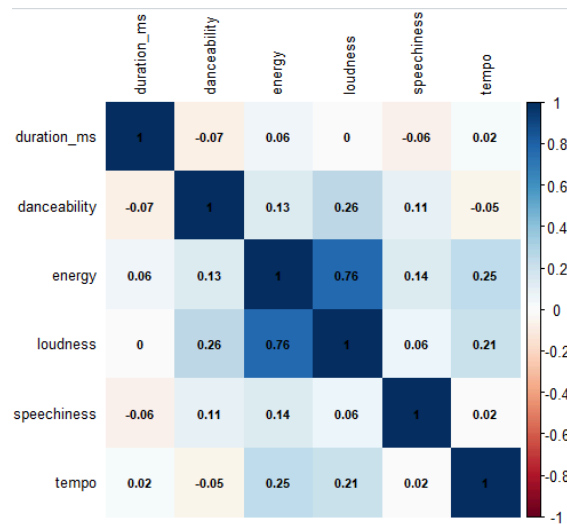
```

Response: popularity
      Sum Sq   Df F value    Pr(>F)
newduration 19005   1  51.3    8e-13 ***
danceability 28650   1  77.3 < 2e-16 ***
newenergy   78771   1 212.6 < 2e-16 ***
newloudness 48244   1 130.2 < 2e-16 ***
newspeechiness 5567   1  15.0 0.00011 ***
tempo       4944   1  13.3 0.00026 ***
genre      13825519 113 330.3 < 2e-16 ***
Residuals  42126752 113721
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Correlation matrix:

(3)

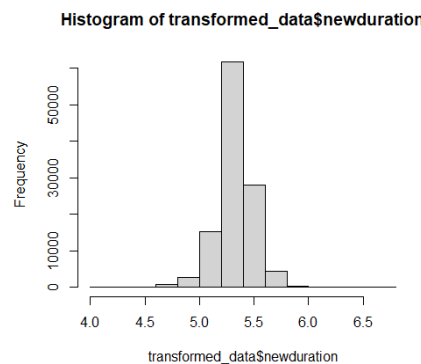
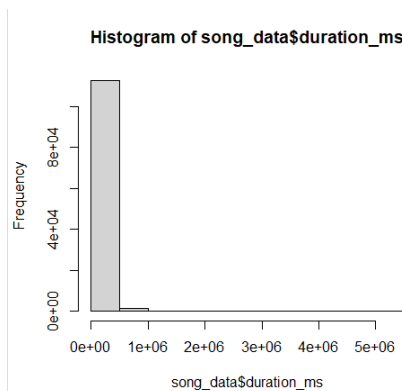


Transformation:

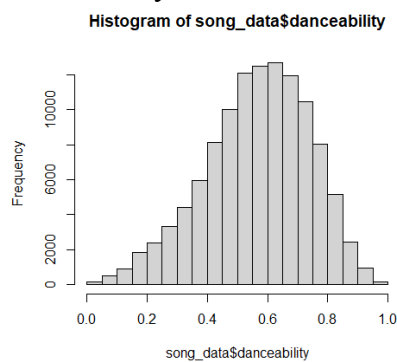
For the following pictures, the left image is before transformation, right image is after transformation:

(4)

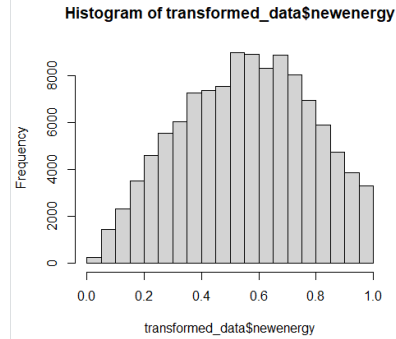
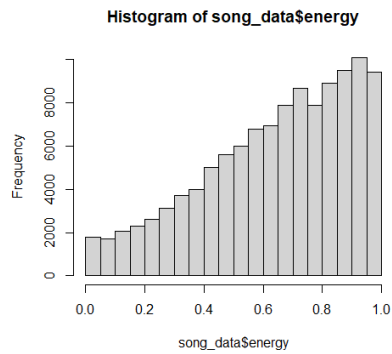
Duration:



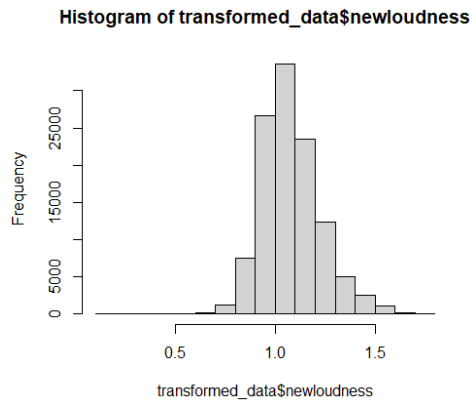
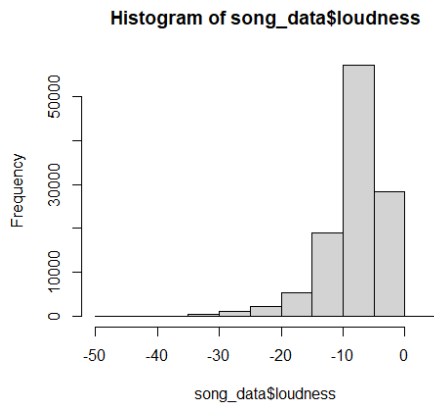
Danceability:



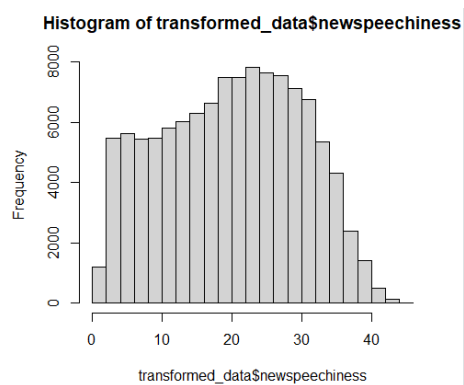
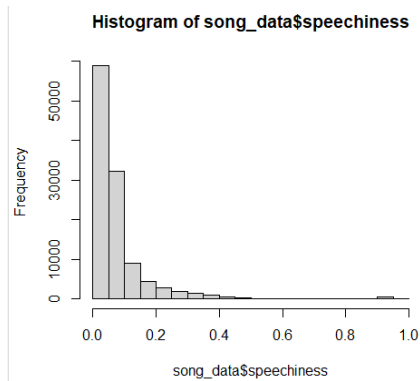
Energy:



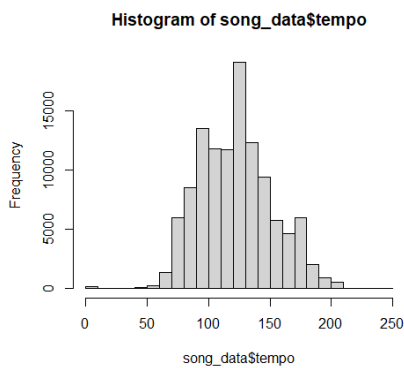
Loudness:



Speechiness:



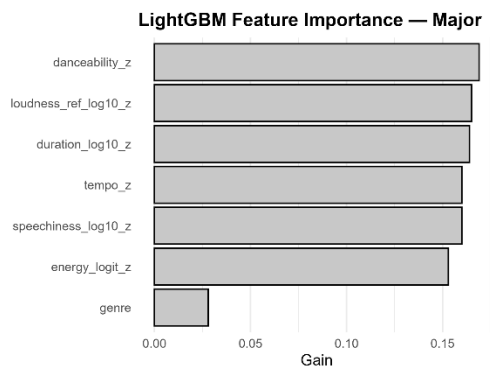
Tempo:



Feature Importance (LightGBM Model):
(5)

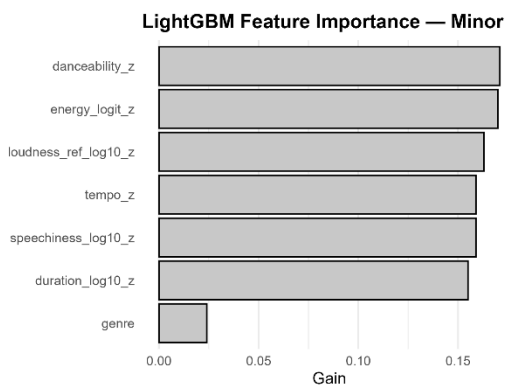
Mode:

Major:



Feature	Gain	Cover	Frequency
danceability_z	0.169	0.159	0.171
loudness_ref_log10_z	0.165	0.150	0.163
duration_log10_z	0.164	0.154	0.166
tempo_z	0.160	0.155	0.166
speechiness_log10_z	0.160	0.151	0.165
energy_logit_z	0.153	0.156	0.158
genre	0.028	0.075	0.012

Minor:

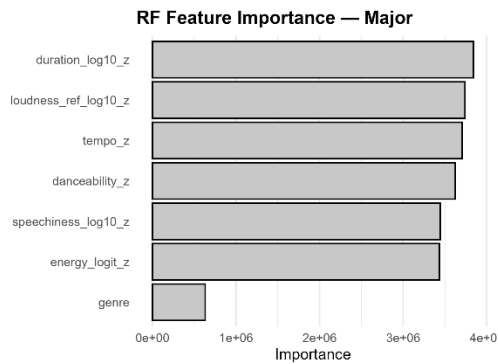


Feature	Gain	Cover	Frequency
danceability_z	0.171	0.157	0.171
energy_logit_z	0.170	0.159	0.164
loudness_ref_log10_z	0.163	0.157	0.161
tempo_z	0.159	0.175	0.163
speechiness_log10_z	0.159	0.149	0.169
duration_log10_z	0.155	0.149	0.162
genre	0.024	0.054	0.009

Feature Importance - Random Forest Analysis:

(6)

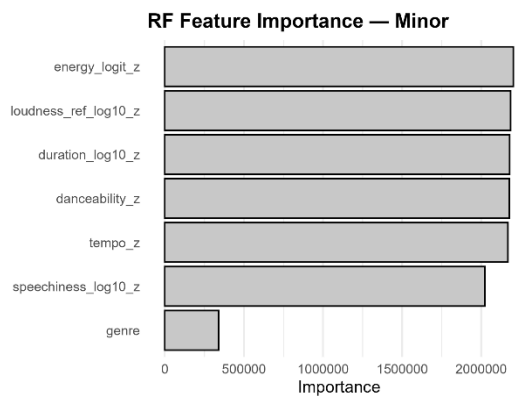
Major:



Feature	Importance
duration_log10_z	3839759.0
loudness_ref_log10_z	3738881.7
tempo_z	3706087.1
danceability_z	3623308.7
speechiness_log10_z	3444633.7
energy_logit_z	3433608.8
genre	630720.8

Minor:

(7)

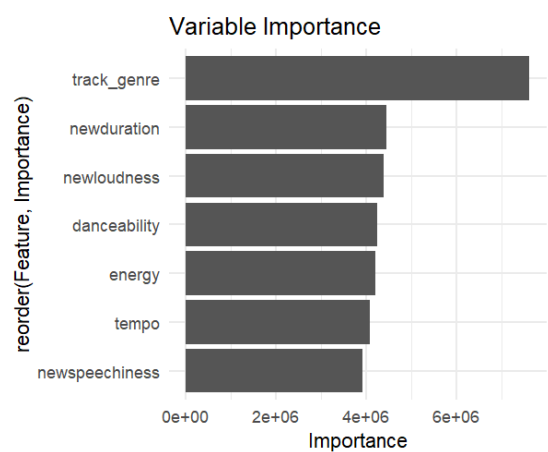


Feature	Importance
energy_logit_z	2203005.9
loudness_ref_log10_z	2185574.6
duration_log10_z	2179864.1
danceability_z	2178044.7
tempo_z	2168724.3
speechiness_log10_z	2023680.8
genre	340355.4

Feature Importance (LightGBM Model):

Explicit:

(8)

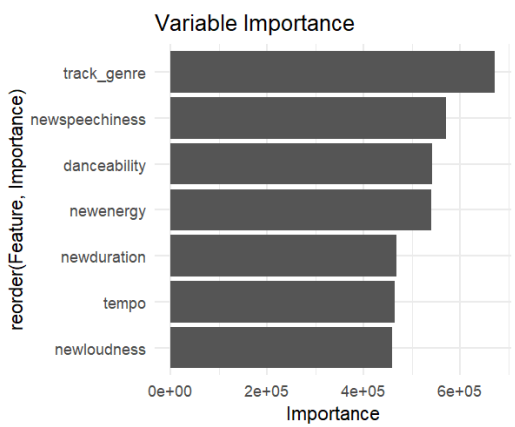


		Feature Importance
newduration	newduration	4435366
danceability	danceability	4232168
newspeechiness	newspeechiness	3906900
energy	energy	4202853
tempo	tempo	4080543
newloudness	newloudness	4379617
track_genre	track_genre	7607240

Non-Explicit:

Feature Importance - Random Forest Analysis:

(9)

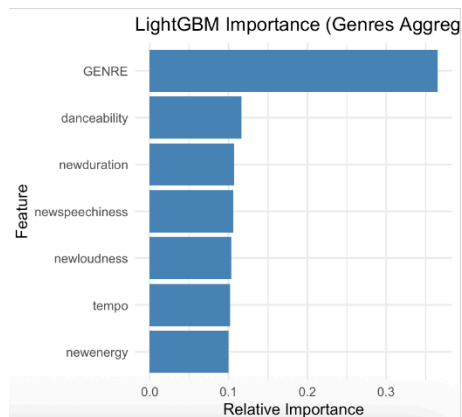


		Feature Importance
newduration	newduration	467010.9
danceability	danceability	541214.7
newspeechiness	newspeechiness	570040.8
newenergy	newenergy	539954.7
tempo	tempo	464493.0
newloudness	newloudness	459454.3
track_genre	track_genre	671140.2

Feature Importance (LightGBM Model):

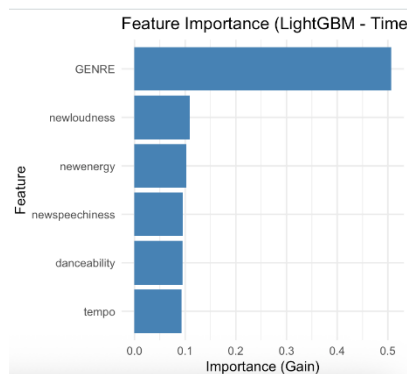
(10)

Time Signature 3:



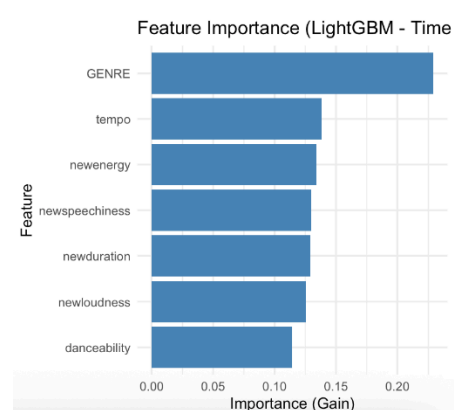
	Variable	Importance
2	GENRE	0.3652359
1	danceability	0.1165260
3	newduration	0.1067728
6	newspeechiness	0.1061813
5	newloudness	0.1033882
7	tempo	0.1018939
4	newenergy	0.1000019

Time Signature 4:



	Variable	Importance
2	GENRE	0.2290534
7	tempo	0.1381967
4	newenergy	0.1342607
6	newspeechiness	0.1296919
3	newduration	0.1287776
5	newloudness	0.1255570
1	danceability	0.1144628

Time Signature 5:

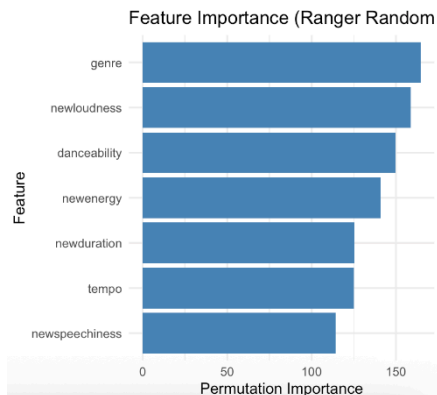


	Variable	Importance
2	GENRE	0.50623571
4	newloudness	0.10874607
3	newenergy	0.10191643
5	newspeechiness	0.09577610
1	danceability	0.09465817
6	tempo	0.09266752

Feature Importance - Random Forest Analysis:

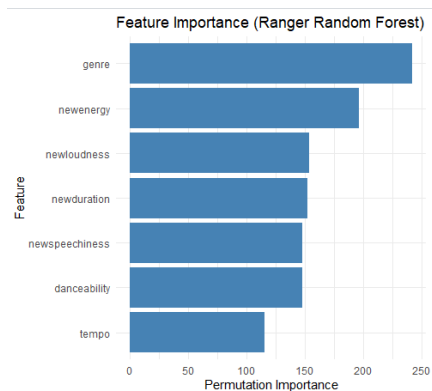
(11)

Time Signature 3:



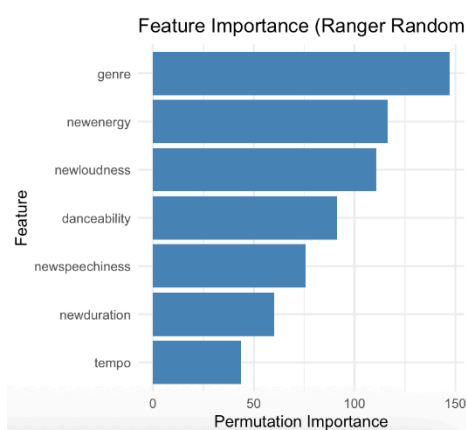
Variable		Importance
genre	genre	164.5067
newloudness	newloudness	158.8619
danceability	danceability	149.8939
newenergy	newenergy	140.8816
newduration	newduration	125.4763
tempo	tempo	125.1648
newspeechiness	newspeechiness	114.2397

Time Signature 4:



Variable		Importance
genre	genre	241.6703
newenergy	newenergy	196.5171
newloudness	newloudness	153.7734
newduration	newduration	152.1428
newspeechiness	newspeechiness	147.9784
danceability	danceability	147.4912
tempo	tempo	115.5428

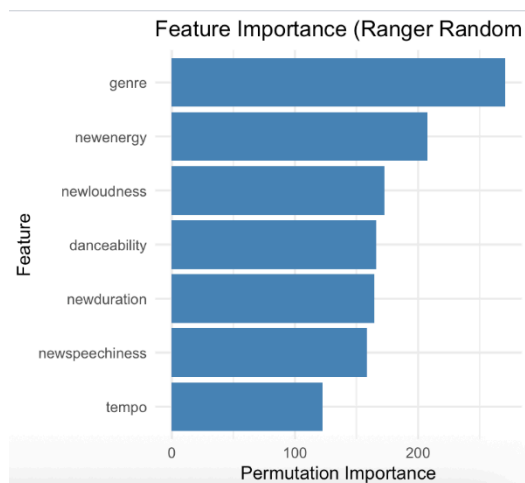
Time Signature 5:



Variable		Importance
genre	genre	147.06488
newenergy	newenergy	116.60497
newloudness	newloudness	110.79491
danceability	danceability	91.40597
newspeechiness	newspeechiness	75.66406
newduration	newduration	60.21993
tempo	tempo	43.64314

Feature Importance – Overall model:

(12)



Variable Importance	
genre	270.5107
newenergy	207.6476
newloudness	172.7363
danceability	166.1365
newduration	164.2098
newspeechiness	158.3940
tempo	122.2572

Predicted vs Observed popularity – overall model:

(13)

